Machine Learning — PYQ Topic Analysis

SPPU B.E. (AI & DS) 2019 Pattern · Course 417521 · 70 marks, 2.5 hours · Each paper: 8 questions in 4 OR-pairs; one pair per unit (Units III–VI of the syllabus, numbered 1–4 here).

1. Papers Analysed

#	Session	Paper code	File	Sub-questions
1	Nov-Dec 2023	[6181]-404, P6779	PYQ/Machine Learning/Nov_Dec_2023.pdf	19
2	Nov-Dec 2024	[6354]-781, PC2632	PYQ/Machine Learning/Nov_Dec_2024.pdf	17
3	Nov-Dec 2025	[6584]-383, PE2433	PYQ/Machine Learning/Nov_Dec_2025.pdf	20
	Total			56

2. Unit-wise Canonical Questions

Importance rule used: ★★★ = asked in all 3 papers, or in 2 papers with ≥ 8 marks; ★★ = asked in 2 papers (< 8 marks); ★ = asked once.

Unit 1: Supervised Learning: Classification & Regression (Syllabus Unit III)

No.	Canonical question	Max M	Years asked	Freq.	Imp.
U1-1	Explain the K-Nearest Neighbour (KNN) classification algorithm with a suitable example (including numerical).	12	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U1-2	Evaluation measures of binary classification: accuracy, precision, recall, F1-score (formula + numerical on confusion matrix).	12	Nov-Dec 2023, Nov-Dec 2024	2	★★★
U1-3	What are kernel functions in SVM? Describe Linear, Polynomial, Radial Basis (Gaussian) and Sigmoid kernels.	6	Nov-Dec 2024, Nov-Dec 2025	2	★★
U1-4	Differentiate between Balanced and Imbalanced classification.	6	Nov-Dec 2024, Nov-Dec 2025	2	★★
U1-5	Explain the Support Vector Machine (SVM) classification algorithm with a suitable example.	6	Nov-Dec 2023	1	★
U1-6	Explain the concept of soft margin SVM. How does it differ from hard margin SVM?	6	Nov-Dec 2025	1	★
U1-7	Explain construction of a multi-class classifier: (i) One-vs-All (ii) One-vs-One.	6	Nov-Dec 2023	1	★
U1-8	Differentiate between Binary and Multiclass classification.	6	Nov-Dec 2023	1	★
U1-9	Explain any 4 evaluation measures of Multiclass classification.	4	Nov-Dec 2024	1	★
U1-10	Explain evaluation measures (SSE, MSE) for a regression model, with numerical.	6	Nov-Dec 2025	1	★
U1-11	Explain Bayesian Linear Regression.	6	Nov-Dec 2025	1	★

Raw questions mapped (exact wording):

- Nov-Dec 2023 Q1(a) [12] → **U1-1**: Apply K-Nearest Neighbor Algorithm (KNN) on following data. Predict the student result for values physics = 6 marks, Chemistry = 8 marks. Consider number of neighbours K = 3 and Euclidean Distance as distance measure.
- Nov-Dec 2024 Q2(c) [8] → **U1-1**: Write a detailed note on K Nearest Neighbour algorithm with suitable example.
- Nov-Dec 2025 Q1(b) [6] → **U1-1**: Explain K nearest neighbor classification algorithm with suitable example.
- Nov-Dec 2023 Q2(a) [6] → **U1-2**: Explain any 4 evaluation measures of Binary classification with example?
- Nov-Dec 2024 Q1(b) [12] → **U1-2**: Write formula for accuracy, precision, recall and f1-score. Calculate accuracy, precision, recall and f1-score for given example. (Confusion matrix: Predicted Cancer -> 45 actual Cancer, 18 actual No Cancer; Predicted No Cancer -> 12 actual Cancer, 25 actual No Cancer)
- Nov-Dec 2024 Q1(a) [6] → **U1-3**: What are kernel functions in SVM? Describe the Radial Basis Kernel, and Sigmoid kernel.
- Nov-Dec 2025 Q1(c) [6] → **U1-3**: What are kernel functions in SVM? Describe the Radial Basis Kernel, Gaussian, Polynomial, and Sigmoid kernel

- Nov-Dec 2024 Q2(b) [6] → **U1-4**: Differentiate Balanced and Imbalanced Classification.
- Nov-Dec 2025 Q2(b) [6] → **U1-4**: Differentiate between balanced and imbalanced classification.
- Nov-Dec 2023 Q1(b) [6] → **U1-5**: Explain support Vector Machine classification algorithm with suitable example.
- Nov-Dec 2025 Q2(c) [6] → **U1-6**: Explain the concept of a soft margin SVM. How does it differ from a hard margin SVM?
- Nov-Dec 2023 Q2(b) [6] → **U1-7**: Explain construction of multi-classifier. i) One Vs. All approach ii) One Vs One approach
- Nov-Dec 2023 Q2(c) [6] → **U1-8**: Differentiate between Binary - vs - Multiclass Classification.
- Nov-Dec 2024 Q2(a) [4] → **U1-9**: Explain any 4 evaluation measures of Multiclass classification.
- Nov-Dec 2025 Q1(a) [6] → **U1-10**: Explain evaluation measures (SSE, MSE) for regression model. For a given data having 100 examples, if squared errors SE1, SE2, and SE3 are 13.33, 3.33 and 4.00 respectively, calculate Mean Squared Error (MSE). State the formula for MSE.
- Nov-Dec 2025 Q2(a) [6] → **U1-11**: Explain Baysian Linear Regression.

Unit 2: Unsupervised Learning: Clustering (Syllabus Unit IV)

No.	Canonical question	Max M	Years asked	Freq.	Imp.
U2-1	Explain the K-Means (centroid-based) clustering algorithm with example. State its advantages and disadvantages.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U2-2	Explain the DBSCAN algorithm. How does it work? State its advantages and disadvantages.	9	Nov-Dec 2024, Nov-Dec 2025	2	★★★
U2-3	Describe the Gaussian Mixture Model (GMM) for distribution-based clustering with example. How is it different from other clustering algorithms?	9	Nov-Dec 2023, Nov-Dec 2025	2	★★★
U2-4	What is clustering? Elaborate the types of clustering.	9	Nov-Dec 2024	1	★
U2-5	Elaborate the need of clustering and explain how the elbow method is used to decide the value of k.	9	Nov-Dec 2023	1	★
U2-6	Differentiate Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering.	8	Nov-Dec 2024	1	★
U2-7	Explain the Divisive Hierarchical Clustering (DHC) algorithm with example.	8	Nov-Dec 2023	1	★
U2-8	Discuss applications of clustering in market segmentation, social network analysis, image segmentation and anomaly detection, with examples.	8	Nov-Dec 2025	1	★

Raw questions mapped (exact wording):

- Nov-Dec 2023 Q3(a) [9] → **U2-1**: Explain K - Means clustering algorithm and states the advantages and disadvantages of k-means clustering algorithm.
- Nov-Dec 2024 Q4(b) [9] → **U2-1**: Describe centroid based clustering algorithm and explain any one type with example.
- Nov-Dec 2025 Q4(b) [8] → **U2-1**: Explain the K-Means clustering algorithm. What are its advantages and disadvantages.
- Nov-Dec 2024 Q4(a) [8] → **U2-2**: Explain DBSCAN algorithm with advantages and disadvantages.
- Nov-Dec 2025 Q3(a) [9] → **U2-2**: Explain the DBSCAN algorithm. How does it work? What are its advantages and disadvantages.
- Nov-Dec 2023 Q3(b) [8] → **U2-3**: Explain Gaussian mixture model with example.
- Nov-Dec 2025 Q4(a) [9] → **U2-3**: Describe the Gaussian Mixture Model (GMM) for distribution-based clustering. How is it different from other clustering algorithms?
- Nov-Dec 2024 Q3(b) [9] → **U2-4**: What is clustering? Elaborate Types of Clustering.
- Nov-Dec 2023 Q4(a) [9] → **U2-5**: Elaborate need of clustering and explain how the elbow method is used to decide the value of cluster k.
- Nov-Dec 2024 Q3(a) [8] → **U2-6**: Differentiate Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering.
- Nov-Dec 2023 Q4(b) [8] → **U2-7**: Explain Divisive Hierarchical clustering (DHC) algorithm with example.
- Nov-Dec 2025 Q3(b) [8] → **U2-8**: Discuss the applications of clustering techniques in market segmentaion, social network analysis, image segmentation, and anomaly detection. Provide examples for each application.

Unit 3: Ensemble Learning (Syllabus Unit V)

No.	Canonical question	Max M	Years asked	Freq.	Imp.

U3-1	What is ensemble learning? Explain its need, advantages and limitations, and any two ensemble techniques.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U3-2	Explain the voting ensemble and the different types of voting mechanisms.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U3-3	Explain the stacking approach of ensemble learning with example. What methods are used for variance reduction in stacking?	9	Nov-Dec 2024, Nov-Dec 2025	2	★★★
U3-4	Explain the Adaptive Boosting (AdaBoost) algorithm in detail.	6	Nov-Dec 2023, Nov-Dec 2025	2	★★
U3-5	Explain the Random Forest ensemble method with an example.	6	Nov-Dec 2023, Nov-Dec 2025	2	★★
U3-6	Differentiate between Homogeneous and Heterogeneous ensemble methods, with examples.	6	Nov-Dec 2023, Nov-Dec 2025	2	★★
U3-7	Explain Gradient Boosting ensemble learning technique.	9	Nov-Dec 2024	1	★
U3-8	Differentiate the Bagging and Boosting approaches of ensemble learning.	6	Nov-Dec 2023	1	★

Raw questions mapped (exact wording):

- Nov-Dec 2023 Q6(b) [6] → **U3-1**: What is the ensemble learning? Explain any two ensemble learning techniques.
- Nov-Dec 2024 Q5(a) [9] → **U3-1**: Explain Following with respect to Ensemble learning. i) Need of Ensemble Learning ii) Advantages of Ensemble methods iii) Ensemble learning limitations
- Nov-Dec 2025 Q5(a) [6] → **U3-1**: Explain the concept of ensemble learning. Why is ensemble learning considered beneficial in machine learning?
- Nov-Dec 2023 Q5(b) [6] → **U3-2**: Explain different types of voting mechanisms in ensemble learning.
- Nov-Dec 2024 Q6(b) [9] → **U3-2**: Explain any three voting mechanism in ensemble learning.
- Nov-Dec 2025 Q6(b) [6] → **U3-2**: Describe the concept of a voting ensemble. What are the different types of voting techniques?
- Nov-Dec 2024 Q5(b) [9] → **U3-3**: Elaborate stacking approach of ensemble with example.
- Nov-Dec 2025 Q5(b) [6] → **U3-3**: Discuss the concept of stacking in ensemble learning. What are the different methods used for variance reduction in stacking?
- Nov-Dec 2023 Q5(c) [6] → **U3-4**: Explain AdaBoost algorithm in detail.
- Nov-Dec 2025 Q6(c) [6] → **U3-4**: Explain the Adaptive Boosting (AdaBoost) algorithm in detail.
- Nov-Dec 2023 Q6(c) [6] → **U3-5**: Explain random forest ensembles with an example.
- Nov-Dec 2025 Q5(c) [6] → **U3-5**: Describe the Random Forest ensemble method in detail.
- Nov-Dec 2023 Q6(a) [6] → **U3-6**: Compare Homogeneous and Heterogeneous ensemble methods.
- Nov-Dec 2025 Q6(a) [6] → **U3-6**: Differentiate between homogeneous and heterogeneous ensemble methods. provide examples of each type.
- Nov-Dec 2024 Q6(a) [9] → **U3-7**: Describe ensemble learning. Explain Gradient Boosting ensemble learning techniques.
- Nov-Dec 2023 Q5(a) [6] → **U3-8**: Differentiate the Bagging and Boosting approach of ensemble learning.

Unit 4: Reinforcement Learning (Syllabus Unit VI)

No.	Canonical question	Max M	Years asked	Freq.	Imp.
U4-1	What is Reinforcement Learning? Explain its need/importance, types and real-time applications.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U4-2	Compare supervised, unsupervised and reinforcement learning with examples.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	3	★★★
U4-3	Explain the Q-Learning algorithm: important terms, working, and an example.	9	Nov-Dec 2023, Nov-Dec 2025	2	★★★
U4-4	Explain the Markov property and Markov Decision Process (MDP). How does the Markov property relate to RL and why is it important?	9	Nov-Dec 2024, Nov-Dec 2025	2	★★★
U4-5	Explain the terms: Markov Property, Markov Chain, Markov Reward Process, Bellman Equation, Q-table, Q-function.	8	Nov-Dec 2023, Nov-Dec 2024	2	★★★

Raw questions mapped (exact wording):

- Nov-Dec 2023 Q8(a) [8] → **U4-1**: What is Reinforcement Learning? Explain the real time applications of reinforcement learning.
- Nov-Dec 2024 Q7(b) [9] → **U4-1**: Explain Reinforcement Learning need and its types in detail?
- Nov-Dec 2025 Q7(a) [9] → **U4-1**: Define reinforcement learning. Why is reinforcement learning important in the field of machine learning?
- Nov-Dec 2023 Q8(b) [9] → **U4-2**: Explain following terms: i) Supervised Learning. ii) Unsupervised Learning. iii) Reinforcement Learning.
- Nov-Dec 2024 Q7(a) [8] → **U4-2**: Differentiate supervised and unsupervised learning with example.
- Nov-Dec 2025 Q8(b) [8] → **U4-2**: Compare and contrast supervised, unsupervised, and reinforcement learning. Provide examples of each type.
- Nov-Dec 2023 Q7(b) [9] → **U4-3**: Explain Q-Learning algorithm with an example.
- Nov-Dec 2025 Q8(a) [9] → **U4-3**: Introduce Q-learning. What are the important terms used in Q-learning? How does Q-learning work?
- Nov-Dec 2024 Q8(b) [9] → **U4-4**: How does the Markov property relate to Reinforcement Learning? Why is it important?
- Nov-Dec 2025 Q7(b) [8] → **U4-4**: Explain Markov's Decision Process (MDP). What is the Markov property in the context of MDP?
- Nov-Dec 2023 Q7(a) [8] → **U4-5**: Explain following terms: i) Markov Property ii) Bellman Equation iii) Markov Reward Process iv) Markov Chain
- Nov-Dec 2024 Q8(a) [8] → **U4-5**: Explain following terms: i) Belman Equation ii) Markov Chain iii) Q table iv) Q function

3. Marks Weightage per Unit

Unit	Name	Nov-Dec 2023 (both OR options)	Nov-Dec 2024 (both OR options)	Nov-Dec 2025 (both OR options)	Marks per paper (attempted)	Canonical Qs	★★★
1	Supervised Learning: Classification & Regression	36	36	36	17–18	11	2
2	Unsupervised Learning: Clustering	34	34	34	17–18	8	3
3	Ensemble Learning	36	36	36	17–18	8	3
4	Reinforcement Learning	34	34	34	17–18	5	5

Each OR-pair carries 17 or 18 marks (Q1/Q2 = 18, Q3/Q4 = 17, Q5/Q6 = 18, Q7/Q8 = 17); the per-year columns add both OR options, so they are about double the attempted marks.

4. Top 10 Most Important Topics

Ranked by frequency first, then by marks. Each was asked in **all 3 papers** or in **2 papers for 8–12 marks**.

Rank	Topic	Unit	Asked in	Max marks	Why it matters
1	**KNN algorithm + numerical**	1	2023, 2024, 2025	12	Every paper; 12-mark numerical in 2023
2	**K-Means clustering (centroid-based) + pros/cons**	2	2023, 2024, 2025	9	Every paper, always 8–9 marks
3	**Ensemble learning: concept, need, advantages, limitations**	3	2023, 2024, 2025	9	Every paper
4	**Voting ensemble and types of voting**	3	2023, 2024, 2025	9	Every paper
5	**Reinforcement learning: definition, need, types, applications**	4	2023, 2024, 2025	9	Every paper
6	**Supervised vs Unsupervised vs Reinforcement learning**	4	2023, 2024, 2025	9	Every paper
7	**Confusion matrix: accuracy, precision, recall, F1 (numerical)**	1	2023, 2024	12	12-mark numerical
8	**Q-Learning (terms, update rule, example)**	4	2023, 2025	9	9 marks both times
9	**Markov property / MDP + RL terms (Markov chain, MRP, Bellman, Q-table, Q-function)**	4	2023, 2024, 2025	9	Asked in some form every year (U4-4 + U4-5)

10	**DBSCAN and GMM (density and distribution clustering)**	2	2023, 2024, 2025	9	Each asked twice at 8–9 marks

Next in line: stacking (2024, 2025; 9 M), SVM kernels (2024, 2025), balanced vs imbalanced (2024, 2025), AdaBoost, Random Forest, Homogeneous vs Heterogeneous (2023, 2025).

Strategy: Unit 4 (RL) is the most predictable: 5 canonical questions, all ★★★, and they repeat every year. Unit 1 always has one 12-mark numerical (KNN or confusion matrix), so practise both.

5. Assumptions & Gaps

- **Papers available:** only 3 ML papers (Nov-Dec 2023, 2024, 2025). With 3 papers, "more than half" means 2 or more papers, which would leave ★★ empty. So the rule used is: ★★★ = all 3 papers, or 2 papers with max marks ≥ 8; ★★ = 2 papers with < 8 marks; ★ = once.
- **Unit labels:** the papers do not name units. Mapping used: Q1/Q2 → Unit 1 (Syllabus III, Classification), Q3/Q4 → Unit 2 (IV, Clustering), Q5/Q6 → Unit 3 (V, Ensemble), Q7/Q8 → Unit 4 (VI, Reinforcement Learning). This matches the chapter order of ML Unit 3-6.pdf.
- **Merged questions:** "centroid-based clustering, explain one type" (2024) merged with K-Means; "what is ensemble + two techniques" (2023) merged with ensemble concept/need; the RL term lists (2023, 2024) merged into one canonical question; supervised vs unsupervised (2024) merged with the 3-way comparison.
- **Ambiguous data:** 2024 Q1(b) confusion matrix: rows were read as *Predicted* and columns as *Actual* (as printed). 2025 Q1(a) MSE wording is unclear; both readings are solved in the Unit 1 document.
- **Materials:** no separate lecture notes were found. Sources are the two scanned textbooks (ML Unit 3-6.pdf and ML Textbook.pdf), OCR'd. Page numbers are **PDF page numbers**.
- ⚠ **Topics not found in your materials:** U3-7 Gradient Boosting (syllabus listing only).

6. Coverage Check

Raw sub-questions extracted: **56** (2023: 19, 2024: 17, 2025: 20). Unique (paper, question) keys: **56** → every raw question maps to exactly one canonical question. Canonical questions: **32**.